

# CENOBOTS OPEN API

## Update Log

### V1.0.16

1. Supplemented descriptions for specific fields.(2026-04-16)

### V1.0.15

New features(2026-04-08)

1. Added `Get robot info` API endpoint
2. Added `Get robot's maintenance detail` API endpoint
3. Added query parameter `isShowNavigationMission` and response fields `missionInitiator` and `missionCanceller` to the `Query history missions by page` API endpoint

### V1.0.14

New features(2026-03-17)

1. `Get device status` returns `remark`

### V1.0.13

New features(2026-03-02)

1. `Query history missions by page` returns `taskEndReasonEnum`

### V1.0.12

New features(2026-02-10)

1. Add a new map interface: Get screenshot of map with four coordinate points

## V1.0.11

New features(2025-11-13)

1. `query history missions by page` returns `startBattery` , `endBattery` , `taskSourceTypeEnum` , `meanVel` and `cumulativeDistance` info

## V1.0.10

New features(2025-10-16)

Add a new device interface: List device system errors

## V1.0.9

New features(2025-08-28)

1. New Interfaces of favorite:
  1. Create new favorite mission
  2. Update favorite mission
  3. Delete favorite msision
  4. Run favorite mission
  5. List favorite missions
2. New Interfaces of robot
  1. Get robot setting
  2. Update robot setting

## V1.0.8

New features(2025-05-06)

1. Add more details in device status
  1. Current area ids of running task
  2. Current lap of running task
2. New type when create task:
  1. Strict navigation task which can constraint angle

## V1.0.7

Add audio and map open api(2024-12-20)

1. New interfaces of Audio:
  1. Get volumn from device

2. Set device volumn
  3. Play audio on device
  4. Stop play audio
2. New interfaces of Map:
1. List all maps of device
  2. Get screenshot of map

## V1.0.6

1. Add example code for python and java(2024-11-21)

## V1.0.5

Add some task end back point interfaces (2024-08-26)

1. New interfaces:
  1. Create back point
  2. Update back point
  3. List back points
  4. Delete back point
2. `Create new temporary mission` and `Create new schedule` add new parameter `backPointId` to specify where the robot will go when the task is finished

## V1.0.4

1. `Query history missions by page` return 'taskSummaryPictureUrl' as the picture url of task summary

## V1.0.3

1. `Query history missions by page` return 'errDesc' when the task is abnormal end

## V1.0.2

1. `Create new temporary/navigate mission` return mission id if created successfully
2. `Query history missions by page` return mission start timestamp and end timestamp
3. `Get device status` return current mission id when device is running

**Description:**Device Open API

**HOST:** http://{Host}/app

**Version:** v1.0.8

## CENOBOTS Open API

### Update Log

V1.0.16  
V1.0.15  
V1.0.14  
V1.0.13  
V1.0.12  
V1.0.11  
V1.0.10  
V1.0.9  
V1.0.8  
V1.0.7  
V1.0.6  
V1.0.5  
V1.0.4  
V1.0.3  
V1.0.2

### Example Code

Java

Python

### Schedule

Create new schedule

Update schedule

Disable-Enable schedule

Get all schedule by device

Delete schedule

### Audio

Set device volume

Stop play audio on device

Start play audio on device

Get current volume from device

### Mission

Create new temporary mission

Get mission summary

Navigate to (x, y, z, yaw) in the given map which can constraint angle

Navigate to (x, y, z, yaw) in the given map

Query history missions by page

Go home  
Stop current mission  
Pause current mission  
Continue current mission

### Map

Change map  
List all areas by device  
Create back point  
Update back point  
List Back points  
Delete Back Point  
List all current maps of device  
Get screenshot of map  
Get screenshot of map with four coordinate points

### Favorite

Create new favorite mission  
Update favorite mission  
Delete favorite mission  
Run favorite mission  
List favorite missions

### Device

Get device status  
Get robot info  
Get robot's maintenance detail  
Get robot setting  
Update robot setting  
List device system errors  
Get device open-id list

EXAMPLE CODE

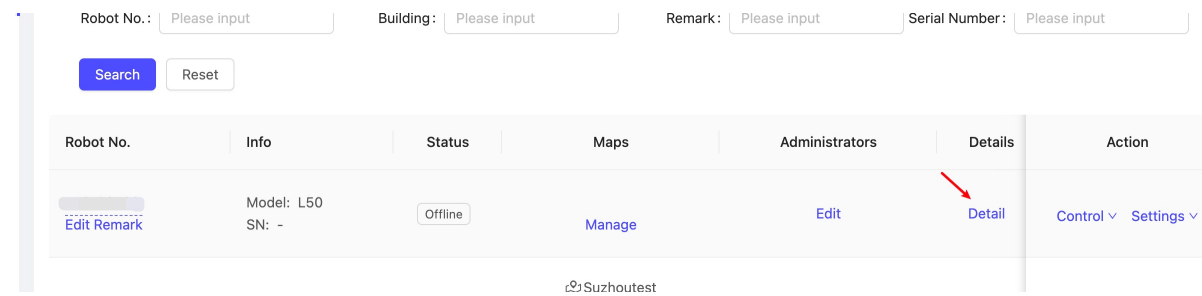
Notice:

1. The `HOST` varies in different regions

1. APAC use <https://app-server-apac.cz-robots.com>
2. EU use <https://app-server-eu.cz-robots.com>
3. US use <https://app-server-us.cz-robots.com>
4. Other use <https://app-server.cz-robots.com>

2. Get the robot's openId

View <https://operation.cz-robots.com/robots> and click the 'Detail' button



## JAVA

```
package com.czrobots.example;

import com.alibaba.fastjson2.JSON;

import javax.crypto.Mac;
import javax.crypto.spec.SecretKeySpec;
import java.io.BufferedReader;
import java.io.DataOutputStream;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
import java.nio.charset.StandardCharsets;
import java.time.Instant;
import java.util.HashMap;
import java.util.Map;

public class OpenApiExample {

    enum Method {
        GET, POST, HEAD, OPTIONS, PUT, DELETE, TRACE, CONNECT, PATCH
    }
}
```

```

private static final String HOST = "https://app-server.cz-robots.com";
private static final String ROBOT_OPEN_ID = "ZXxxxxxx67";
private static final String ACCESS_KEY = "6xxx7d";
private static final String SECRET_KEY = "2xxxbd";

private static final String HEADER_ACCESS_KEY = "X-API-Key";
private static final String HEADER_TIMESTAMP = "X-API-timestamp";
private static final String HEADER_SIGNATURE = "X-API-signature";

private static final char[] alphabets = "0123456789abcdef".toCharArray();

public static String bytesToHex(byte[] data) {
    final int len = data.length;
    final char[] out = new char[len << 1];
    for (int i = 0, j = 0; i < len; i++) {
        out[j++] = alphabets[(0xF0 & data[i]) >>> 4];
        out[j++] = alphabets[0x0F & data[i]];
    }
    return new String(out);
}

public static void getExample() throws Exception {
    String url = "/app/openapi/v1/device/status/" + ROBOT_OPEN_ID;
    String timestamp = Instant.now().toEpochMilli() + "";

    Mac mac = Mac.getInstance("HmacSHA256");
    SecretKeySpec keySpec = new
SecretKeySpec(SECRET_KEY.getBytes(StandardCharsets.UTF_8), "HmacSHA256");
    mac.init(keySpec);
    String signature = bytesToHex(mac.doFinal((Method.GET.name() + timestamp +
url).getBytes(StandardCharsets.UTF_8)));

    HttpURLConnection connection = (HttpURLConnection) new URL(HOST +
url).openConnection();
    connection.setRequestMethod(Method.GET.name());
    connection.setRequestProperty(HEADER_ACCESS_KEY, ACCESS_KEY);
    connection.setRequestProperty(HEADER_TIMESTAMP, timestamp);
    connection.setRequestProperty(HEADER_SIGNATURE, signature);

    int responseCode = connection.getResponseCode();
    System.out.println("Response Code: " + responseCode);
}

```

```

        BufferedReader in = new BufferedReader(new
InputStreamReader(connection.getInputStream()));
        String inputLine;
        StringBuffer content = new StringBuffer();
        while ((inputLine = in.readLine()) != null) {
            content.append(inputLine);
        }
        in.close();
        connection.disconnect();
        System.out.println(content);
    }

    public static void postExample() throws Exception {
        String url = "/app/openapi/v1/mission/list";

        Map<String, Object> dataMap = new HashMap<>();
        dataMap.put("deviceOpenId", ROBOT_OPEN_ID);
        dataMap.put("startTimestamp", "1712725202000");
        dataMap.put("endTimestamp", "1712818791467");

        Map<String, Object> paramMap = new HashMap<>();
        paramMap.put("pageIndex", 1);
        paramMap.put("pageLength", 1);
        paramMap.put("requestData", dataMap);
        String timestamp = Instant.now().toEpochMilli() + "";

        Mac mac = Mac.getInstance("HmacSHA256");
        SecretKeySpec keySpec = new
SecretKeySpec(SECRET_KEY.getBytes(StandardCharsets.UTF_8), "HmacSHA256");
        mac.init(keySpec);
        String signature = bytesToHex(mac.doFinal((Method.POST.name() + timestamp +
url).getBytes(StandardCharsets.UTF_8)));

        HttpURLConnection connection = (HttpURLConnection) new URL(HOST +
url).openConnection();
        connection.setRequestMethod(Method.POST.name());
        connection.setRequestProperty(HEADER_ACCESS_KEY, ACCESS_KEY);
        connection.setRequestProperty(HEADER_TIMESTAMP, timestamp);
        connection.setRequestProperty(HEADER_SIGNATURE, signature);
        connection.setRequestProperty("Content-Type", "application/json");
        connection.setRequestProperty("Accept", "application/json");
        connection.setDoOutput(true);
        try (DataOutputStream wr = new DataOutputStream(connection.getOutputStream())) {
            wr.writeBytes(JSON.toJSONString(paramMap));
            wr.flush();
        }
    }

```

```

        int responseCode = connection.getResponseCode();
        System.out.println("Response Code: " + responseCode);

        BufferedReader in = new BufferedReader(new
InputStreamReader(connection.getInputStream()));
        String inputLine;
        StringBuilder response = new StringBuilder();

        while ((inputLine = in.readLine()) != null) {
            response.append(inputLine);
        }
        in.close();

        System.out.println("Response: " + response);
    }

    public static void main(String[] args) throws Exception {
        getExample();
        postExample();
    }
}

```

## PYTHON

```

import requests
import time
import hmac
import hashlib
import json

HOST = "https://app-server.cz-robots.com"

# Replace these with your actual values.
ROBOT_OPEN_ID = ""
ACCESS_KEY = ""
SECRET_KEY = ""

HEADER_ACCESS_KEY = "X-API-Key"
HEADER_TIMESTAMP = "X-API-timestamp"
HEADER_SIGNATURE = "X-API-signature"

```

```

def bytes_to_hex(data):
    return ''.join(f'{byte:02x}' for byte in data)

def get_example():
    url = f"/app/openapi/v1/device/status/{ROBOT_OPEN_ID}"
    timestamp = str(int(time.time() * 1000))

    signature_string = f"GET{timestamp}{url}".encode('utf-8')
    signature = hmac.new(SECRET_KEY.encode('utf-8'), signature_string,
hashlib.sha256).digest()
    signature_hex = bytes_to_hex(signature)

    headers = {
        HEADER_ACCESS_KEY: ACCESS_KEY,
        HEADER_TIMESTAMP: timestamp,
        HEADER_SIGNATURE: signature_hex
    }

    response = requests.get(HOST + url, headers=headers)
    print("Response Code:", response.status_code)
    print(response.text)

def post_example():
    url = "/app/openapi/v1/mission/list"

    param_map = {
        "pageIndex": 1,
        "pageLength": 1,
        "requestData": {
            "deviceOpenId": ROBOT_OPEN_ID,
            "startTimestamp": "1712725202000",
            "endTimestamp": "1712818791467",
        },
    }

    timestamp = str(int(time.time() * 1000))

    signature_string = f"POST{timestamp}{url}".encode('utf-8')
    signature = hmac.new(SECRET_KEY.encode('utf-8'), signature_string,
hashlib.sha256).digest()
    signature_hex = bytes_to_hex(signature)

    headers = {
        HEADER_ACCESS_KEY: ACCESS_KEY,
        HEADER_TIMESTAMP: timestamp,

```

```
        HEADER_SIGNATURE: signature_hex,
        "Content-Type": "application/json",
        "Accept": "application/json"
    }

    response = requests.post(HOST + url, headers=headers, data=json.dumps(param_map))
    print("Response Code:", response.status_code)
    print("Response:", response.text)

if __name__ == "__main__":
    get_example()
    post_example()
```

## SCHEDULE

### CREATE NEW SCHEDULE

**url:** /app/openapi/v1/schedule

**method:** POST

**produces:** application/x-www-form-urlencoded,application/json

**consumes:** \*/\*

#### Note:

L50 can only create SWEEP and HOMING schedule

If sweepMode is PATROL, subSweepMode can be one of SPOT\_CLEAN,LOW,MEDIUM,HIGH.

If sweepMode is SWEEP, subSweepMode can be one of LOW,MEDIUM,HIGH.

If sweepMode is HOMING, subSweepMode is not necessary and will be ignored.

Example:

```
{
  "duration": 60,
  "mapId": 1100011697639116,
  "mapVersion": "map.1111699509667377.20230616-2.20240115-10",
  "cleanEveryWhere": false,
  "sweepMode": "PATROL",
  "subSweepMode": "HIGH",
  "areaIds": [
    "0",
    "3"
  ],
  "fixedLaps": 1,
  "backPointId": "-1",
  "deviceOpenId": "1234",
  "startTime": "10:40 am",
  "repeat": [
    "Monday",
    "Friday"
  ]
}
```

Params:

| name                      | description                                                                                                                                                                                                                           | in     | require | type           | schema |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|----------------|--------|
| X-API-Key                 | Key                                                                                                                                                                                                                                   | header | true    | string         |        |
| X-API-timestamp           | timestamp                                                                                                                                                                                                                             | header | true    | string         |        |
| X-API-signature           | signature                                                                                                                                                                                                                             | header | true    | string         |        |
| createScheduleOpenRequest | CreateScheduleOpenRequest                                                                                                                                                                                                             | body   | true    |                |        |
| duration                  | task time limit, unit(minutes), (1, 1440). Will be ignored if HOMING                                                                                                                                                                  |        | false   | integer(int32) |        |
| mapId                     | map id that task will use                                                                                                                                                                                                             |        | true    | integer(int64) |        |
| mapVersion                | map id version that task will use                                                                                                                                                                                                     |        | true    | string         |        |
| cleanEveryWhere           | option to enable/disable cleaning of the entire place. if false,arealds should not empty. Will be ignored if HOMING                                                                                                                   |        | false   | boolean        |        |
| sweepMode                 | clean mode,values:PATROL,HOMING,SWEEP                                                                                                                                                                                                 |        | true    | string         |        |
| subSweepMode              | sweep intensity,values:LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                                                                                                     |        | false   | string         |        |
| arealds                   | clean areas, required when cleanEveryWhere is false                                                                                                                                                                                   |        | false   | array          | string |
| fixedLaps                 | fixed laps, required when duration is null, (1,3). Will be ignored if HOMING                                                                                                                                                          |        | false   | integer(int32) |        |
| backPointId               | 1. null or empty string is default behavior go back to station<br>2. -1 represents that device will stop at the end of the task<br>3. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint |        | false   | string         |        |

| name         | description                                            | in | require | type   | schema |
|--------------|--------------------------------------------------------|----|---------|--------|--------|
| deviceOpenId | your device open id                                    |    | true    | string |        |
| startTime    | start time format, eg: '10:40 am'(hours:minutes am/pm) |    | true    | string |        |
| repeat       | repeated days                                          |    | false   | array  | string |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

UPDATE SCHEDULE

url: /app/openapi/v1/schedule

method: PUT

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

L50 can only create SWEEP and HOMING schedule  
If sweepMode is PATROL, subSweepMode can be one of SPOT\_CLEAN,LOW,MEDIUM,HIGH.  
If sweepMode is SWEEP, subSweepMode can bo one of LOW,MEDIUM,HIGH.  
If sweepMode is HOMING, subSweepMode is not necessary and will be ignored.

Example:

```
{
  "duration": 30,
  "mapId": 1100011697639116,
  "mapVersion": "map.1111699509667377.20230616-2.20240115-10",
  "cleanEverywhere": false,
  "sweepMode": "PATROL",
  "subSweepMode": "HIGH",
  "areaIds": [
    "0",
    "3"
  ],
  "fixedLaps": 1,
  "backPointId": "-1",
  "id": 56,
  "deviceOpenId": "1234",
  "startTime": "10:40 am",
  "repeat": [
    "Monday",
    "Friday"
  ]
}
```

Params:

| name                      | description               | in     | require | type           | schema |
|---------------------------|---------------------------|--------|---------|----------------|--------|
| X-API-Key                 | Key                       | header | true    | string         |        |
| X-API-timestamp           | timestamp                 | header | true    | string         |        |
| X-API-signature           | signature                 | header | true    | string         |        |
| updateScheduleOpenRequest | UpdateScheduleOpenRequest | body   | true    |                |        |
| duration                  | task duration             |        | false   | integer(int32) |        |
| mapId                     | map id that task will use |        | true    | integer(int64) |        |

| name            | description                                                                                                                                                                                                                           | in | require | type           | schema |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------|----------------|--------|
| mapVersion      | map id version that task will use                                                                                                                                                                                                     |    | true    | string         |        |
| cleanEveryWhere | option to enable/disable cleaning of the entire place. if false,arealds should not empty. Will be ignored if HOMING                                                                                                                   |    | false   | boolean        |        |
| sweepMode       | clean mode,values:PATROL,HOMING,SWEEP                                                                                                                                                                                                 |    | true    | string         |        |
| subSweepMode    | sweep intensity,values:LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                                                                                                     |    | false   | string         |        |
| arealds         | clean areas, required when cleanEveryWhere is false                                                                                                                                                                                   |    | false   | array          | string |
| fixedLaps       | fixed laps, required when duration is null, (1,3). Will be ignored if HOMING                                                                                                                                                          |    | false   | integer(int32) |        |
| backPointId     | 1. null or empty string is default behavior go back to station<br>2. -1 represents that device will stop at the end of the task<br>3. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint |    | false   | string         |        |
| id              | schedule id                                                                                                                                                                                                                           |    | true    | integer(int64) |        |
| deviceOpenId    | your device open id                                                                                                                                                                                                                   |    | true    | string         |        |
| startTime       | formate start time (hours:minutes am/pm)                                                                                                                                                                                              |    | true    | string         |        |
| repeat          | repeated days                                                                                                                                                                                                                         |    | false   | array          | string |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# DISABLE-ENABLE SCHEDULE

url: /app/openapi/v1/schedule/active

method: POST

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

Note:

Example:

```
{
  "id": 10,
  "deviceId": "1234",
  "enable": true
}
```

Params:

| name                      | description               | in     | require | type                      | schema                    |
|---------------------------|---------------------------|--------|---------|---------------------------|---------------------------|
| X-API-Key                 | Key                       | header | true    | string                    |                           |
| X-API-timestamp           | timestamp                 | header | true    | string                    |                           |
| X-API-signature           | signature                 | header | true    | string                    |                           |
| activeScheduleOpenRequest | ActiveScheduleOpenRequest | body   | true    | ActiveScheduleOpenRequest | ActiveScheduleOpenRequest |
| id                        | schedule id               |        | true    | integer(int64)            |                           |
| deviceId                  | device open id            |        | true    | string                    |                           |
| enable                    | enable/disable schedule   |        | true    | boolean                   |                           |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name | description | type   | schema |
|------|-------------|--------|--------|
| data | dataset     | object |        |

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# GET ALL SCHEDULE BY DEVICE

**url:** /app/openapi/v1/schedule/list/{deviceOpenId}

**method:** GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

Status:

| code | description | schema                           |
|------|-------------|----------------------------------|
| 200  | OK          | ResponseListScheduleTaskOpenResp |

Response Params:

| name            | description                                                                                           | type           | schema               |
|-----------------|-------------------------------------------------------------------------------------------------------|----------------|----------------------|
| data            | dataset                                                                                               | array          | ScheduleTaskOpenResp |
| deviceId        |                                                                                                       | integer(int64) |                      |
| id              | schedule id                                                                                           | integer(int64) |                      |
| duration        | duration (min)                                                                                        | integer(int64) |                      |
| executionDays   | execution days                                                                                        | array          | string               |
| startTime       | start time                                                                                            | string         |                      |
| endTime         | end time                                                                                              | string         |                      |
| mapId           | map id                                                                                                | integer(int64) |                      |
| areas           | list of areas                                                                                         | array          | Area                 |
| id              | user defined area id, required when creating mission/schedule(the name of the parameter is 'arealds') | string         |                      |
| name            | area name                                                                                             | string         |                      |
| type            | area type                                                                                             | string         |                      |
| sweepMode       | clean mode,values:PATROL,HOMING,SWEEP                                                                 | string         |                      |
| subSweepMode    | sweep intensity,values:LOW,MEDIUM,HIGH,SPOT_CLEAN                                                     | string         |                      |
| cleanEveryWhere | cleaning of the entire map is required                                                                | boolean        |                      |
| enable          | scheduled task effective or not                                                                       | boolean        |                      |
| backPointId     | mission back point id                                                                                 | string         |                      |
| backPointName   | mission back point name                                                                               | string         |                      |
| backPoint       | back point                                                                                            | Point          | Point                |
| x               | x                                                                                                     | number(double) |                      |
| y               | y                                                                                                     | number(double) |                      |
| yaw             | The robot's heading angle in radians. Range: $(-\pi, \pi)$                                            | number(double) |                      |
| fixedLaps       | fixed laps                                                                                            | integer(int32) |                      |
| success         | used to determine whether the call is successful                                                      | boolean        |                      |
| code            | response status code                                                                                  | integer(int32) | integer(int32)       |
| info            | response description                                                                                  | string         |                      |
| rid             | traceld is used to trace the records of API calls                                                     | string         |                      |

Response Example:

```
{
  "deviceId": 0,
  "id": 10,
  "duration": 60,
  "executionDays": [
    "Mon."
  ],
  "startTime": "10:40 am",
  "endTime": "11:40 am",
  "mapId": 1100011697114280,
  "areas": [
    {
      "name": "area name example",
      "id": "705429ee-8aca-4f18-b498-9a458bc18407",
      "type": "divide_area"
    }
  ],
  "sweepMode": "PATROL",
  "subSweepMode": "LOW",
  "cleanEverywhere": true,
  "enable": true,
  "fixedLaps": 1,
  "backPointId": "",
  "backPointName": "",
  "backPoint": {
    "x": 2,
    "y": 2,
    "yaw": 2.1
  }
}
```

## DELETE SCHEDULE

**url:** /app/openapi/v1/schedule/delete/{id}

**method:** DELETE

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type           | schema |
|-----------------|-------------|--------|---------|----------------|--------|
| id              | schedule id | path   | true    | integer(int64) |        |
| X-Api-Key       | Key         | header | true    | string         |        |
| X-Api-timestamp | timestamp   | header | true    | string         |        |
| X-Api-signature | signature   | header | true    | string         |        |

**Status:**

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# AUDIO

## SET DEVICE VOLUME

url: /app/openapi/v1/audio/volume

method: PUT

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

Example:

```
{
  "volume": 50,
  "deviceId": ""
}
```

Params:

| name             | description           | in     | require | type             | schema           |
|------------------|-----------------------|--------|---------|------------------|------------------|
| X-API-Key        | Key                   | header | true    | string           |                  |
| X-API-timestamp  | timestamp             | header | true    | string           |                  |
| X-API-signature  | signature             | header | true    | string           |                  |
| setVolumeRequest | SetVolumeRequest      | body   | true    | SetVolumeRequest | SetVolumeRequest |
| volume           | volume value: 0 - 100 |        | true    | integer(int32)   |                  |
| deviceId         | device open id        |        | true    | string           |                  |

Status:

| code | description | schema |
|------|-------------|--------|
|------|-------------|--------|

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# STOP PLAY AUDIO ON DEVICE

url: /app/openapi/v1/audio/stop

method: PUT

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": ""
}
```

Params:

| name                 | description          | in     | require | type                 | schema               |
|----------------------|----------------------|--------|---------|----------------------|----------------------|
| X-API-Key            | Key                  | header | true    | string               |                      |
| X-API-timestamp      | timestamp            | header | true    | string               |                      |
| X-API-signature      | signature            | header | true    | string               |                      |
| stopPlayAudioRequest | StopPlayAudioRequest | body   | true    | StopPlayAudioRequest | StopPlayAudioRequest |
| deviceOpenId         | device open id       |        | true    | string               |                      |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": "",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# START PLAY AUDIO ON DEVICE

url: /app/openapi/v1/audio/start

method: PUT

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": "",
  "audioName": "001/002/testAudio",
  "selfDefined": false,
  "loop": false,
  "timeout": 0,
  "interval": 0
}
```

Params:

| name             | description                                                                                                         | in     | require | type             | schema           |
|------------------|---------------------------------------------------------------------------------------------------------------------|--------|---------|------------------|------------------|
| X-Api-Key        | Key                                                                                                                 | header | true    | string           |                  |
| X-Api-timestamp  | timestamp                                                                                                           | header | true    | string           |                  |
| X-Api-signature  | signature                                                                                                           | header | true    | string           |                  |
| playAudioRequest | PlayAudioRequest                                                                                                    | body   | true    | PlayAudioRequest | PlayAudioRequest |
| deviceOpenId     | device open id                                                                                                      |        | true    | string           |                  |
| audioName        | system audio name : 001/002..., or self-defined audio name. Please refer to operation platform to obtain audio name |        | true    | string           |                  |
| selfDefined      | play system or self-defined audio                                                                                   |        | true    | boolean          |                  |
| loop             | loop play or not                                                                                                    |        | true    | boolean          |                  |
| timeout          | timeout in seconds, required when play loop                                                                         |        | false   | integer(int32)   |                  |

| name     | description                                                         | in | require | type           | schema |
|----------|---------------------------------------------------------------------|----|---------|----------------|--------|
| interval | interval time in seconds between each play, required when play loop |    | false   | integer(int32) |        |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": "",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# GET CURRENT VOLUME FROM DEVICE

url: /app/openapi/v1/audio/volume/{deviceOpenId}

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

Status:

| code | description | schema          |
|------|-------------|-----------------|
| 200  | OK          | ResponseInteger |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | integer(int32) | integer(int32) |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": 0,
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# MISSION

## CREATE NEW TEMPORARY MISSION

**url:** `/app/openapi/v1/mission`

**method:** `POST`

**produces:** `application/x-www-form-urlencoded,application/json`

**consumes:** `*/*`

### Note:

If cleanEveryWhere is false, userDefinedArea and arealds need at lease one.

L50 can only create SWEEP mission, SP50 can create both PATROL and SWEEP.

If sweepMode is PATROL, subSweepMode can be one of SPOT\_CLEAN,LOW,MEDIUM,HIGH.

If sweepMode is SWEEP, subSweepMode can bo one of LOW,MEDIUM,HIGH.

### Example:

```
{
  "duration": 60,
  "mapId": 1100011697639116,
  "mapVersion": "map.1111699509667377.20230616-2.20240115-10",
  "cleanEverywhere": false,
  "sweepMode": "PATROL",
  "subSweepMode": "HIGH",
  "areaIds": [
    "0",
    "3"
  ],
  "fixedLaps": 1,
  "backPointId": "-1"
  "deviceOpenId": "1234",
```

```
    "userDefinedArea": [
      {
        "x": 0,
        "y": 1
      },
      {
        "x": 0,
        "y": 2
      },
      {
        "x": 1,
        "y": 1
      }
    ]
  }
}
```

Params:

| name                        | description                                                                                                                                                                                                                           | in     | require | type           | schema |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|----------------|--------|
| X-Api-Key                   | Key                                                                                                                                                                                                                                   | header | true    | string         |        |
| X-Api-timestamp             | timestamp                                                                                                                                                                                                                             | header | true    | string         |        |
| X-Api-signature             | signature                                                                                                                                                                                                                             | header | true    | string         |        |
| createTmpMissionOpenRequest | CreateTmpMissionOpenRequest                                                                                                                                                                                                           | body   | true    |                |        |
| duration                    | task time limit, unit(minutes), (1, 1440).                                                                                                                                                                                            |        | false   | integer(int32) |        |
| mapId                       | map id that task will use                                                                                                                                                                                                             |        | true    | integer(int64) |        |
| mapVersion                  | map id version that task will use                                                                                                                                                                                                     |        | true    | string         |        |
| cleanEveryWhere             | option to enable/disable cleaning of the entire place. If false,arealds should not empty.                                                                                                                                             |        | false   | boolean        |        |
| sweepMode                   | clean mode,values:PATROL,SWEEP                                                                                                                                                                                                        |        | true    | string         |        |
| subSweepMode                | sweep intensity,values:LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                                                                                                     |        | false   | string         |        |
| arealds                     | clean areas, required when cleanEveryWhere is false                                                                                                                                                                                   |        | false   | array          | string |
| fixedLaps                   | fixed laps, required when duration is null, (1,3).                                                                                                                                                                                    |        | false   | integer(int32) |        |
| backPointId                 | 1. null or empty string is default behavior go back to station<br>2. -1 represents that device will stop at the end of the task<br>3. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint |        | false   | string         |        |
| deviceOpenId                | your device open id                                                                                                                                                                                                                   |        | true    | string         |        |
| userDefinedArea             | user defined areas                                                                                                                                                                                                                    |        | false   | array          | Point  |
| x                           | x                                                                                                                                                                                                                                     |        | false   | number(double) |        |
| y                           | y                                                                                                                                                                                                                                     |        | false   | number(double) |        |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | mission id                                        | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": "1700000000",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# GET MISSION SUMMARY

**url:** /app/openapi/v1/mission/summary

**method:** POST

**produces:** application/x-www-form-urlencoded,application/json

**consumes:** \*/\*

**Note:**

**Example:**

```
{
  "deviceOpenId": "111",
  "startTimestamp": 1712725202000,
  "endTimestamp": 1712725202000
}
```

Params:

| name                    | description                                                                        | in     | require | type                    | schema                  |
|-------------------------|------------------------------------------------------------------------------------|--------|---------|-------------------------|-------------------------|
| X-API-Key               | Key                                                                                | header | true    | string                  |                         |
| X-API-timestamp         | timestamp                                                                          | header | true    | string                  |                         |
| X-API-signature         | signature                                                                          | header | true    | string                  |                         |
| queryMissionOpenRequest | QueryMissionOpenRequest                                                            | body   | true    | QueryMissionOpenRequest | QueryMissionOpenRequest |
| deviceOpenId            | your device open id                                                                |        | true    | string                  |                         |
| startTimestamp          | mission query start timestamp, unit(ms)                                            |        | false   | integer(int64)          |                         |
| endTimestamp            | mission query end timestamp, unit(ms), if it is not set, current date will be used |        | false   | integer(int64)          |                         |

Status:

| code | description | schema                   |
|------|-------------|--------------------------|
| 200  | OK          | ResponseMissionSummaryVO |

Response Params:

| name                  | description                                      | type             | schema           |
|-----------------------|--------------------------------------------------|------------------|------------------|
| data                  |                                                  | MissionSummaryVO | MissionSummaryVO |
| model                 |                                                  | string           |                  |
| totalJobs             | total jobs                                       | integer(int64)   |                  |
| totalWorkingTime      | total working time (min)                         | integer(int64)   |                  |
| totalCleanedArea      | total cleaned area (m²)                          | number(double)   |                  |
| totalWater            | total water (L)                                  | number(double)   |                  |
| totalCleaningSolution | total cleaning solution (ML)                     | number(double)   |                  |
| success               | used to determine whether the call is successful | boolean          |                  |
| code                  | response status code                             | integer(int32)   | integer(int32)   |
| info                  | response description                             | string           |                  |

| name | description                                       | type   | schema |
|------|---------------------------------------------------|--------|--------|
| rid  | traceld is used to trace the records of API calls | string |        |

**Response Example:**

```
{
  "data": {
    "model": "",
    "totalJobs": 0,
    "totalWorkingTime": 0,
    "totalCleanedArea": 0,
    "totalWater": 0,
    "totalCleaningSolution": 0
  },
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

NAVIGATE TO (X, Y, Z, YAW) IN THE GIVEN MAP WHICH CAN  
CONSTRAINT ANGLE

url: /app/openapi/v1/mission/strict\_navigation

method: POST

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": "",
  "point": {
    "x": 0,
    "y": 0,
    "z": 0,
    "yaw": 0
  }
}
```

Params:

| name                  | description                                                | in     | require | type              |
|-----------------------|------------------------------------------------------------|--------|---------|-------------------|
| X-Api-Key             | Key                                                        | header | true    | string            |
| X-Api-timestamp       | timestamp                                                  | header | true    | string            |
| X-Api-signature       | signature                                                  | header | true    | string            |
| navigationOpenRequest | NavigationOpenRequest                                      | body   | true    | NavigationRequest |
| deviceOpenId          | device open id                                             |        | true    | string            |
| point                 | user defined areas                                         |        | true    | Point             |
| x                     | x                                                          |        | false   | number(double)    |
| y                     | y                                                          |        | false   | number(double)    |
| z                     | z                                                          |        | false   | number(double)    |
| yaw                   | The robot's heading angle in radians. Range: $(-\pi, \pi)$ |        | false   | number(double)    |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

Response Params:

| name    | description                                      | type           | schema         |
|---------|--------------------------------------------------|----------------|----------------|
| data    | dataset                                          | string         |                |
| success | used to determine whether the call is successful | boolean        |                |
| code    | response status code                             | integer(int32) | integer(int32) |
| info    | response description                             | string         |                |

| name | description                                       | type   | schema |
|------|---------------------------------------------------|--------|--------|
| rid  | traceld is used to trace the records of API calls | string |        |

**Response Example:**

```
{
  "data": "",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

NAVIGATE TO (X, Y, Z, YAW) IN THE GIVEN MAP

url: /app/openapi/v1/mission/navigation

method: POST

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

**Note:**

**Example:**

```
{
  "deviceOpenId": "",
  "point": {
    "x": 0,
    "y": 0,
    "z": 0,
    "yaw": 0
  }
}
```

**Params:**

| name                  | description                                                | in     | require | type                  |
|-----------------------|------------------------------------------------------------|--------|---------|-----------------------|
| X-API-Key             | Key                                                        | header | true    | string                |
| X-API-timestamp       | timestamp                                                  | header | true    | string                |
| X-API-signature       | signature                                                  | header | true    | string                |
| navigationOpenRequest | NavigationOpenRequest                                      | body   | true    | NavigationOpenRequest |
| deviceOpenId          | device open id                                             |        | true    | string                |
| point                 | user defined areas                                         |        | true    | Point                 |
| x                     | x                                                          |        | false   | number(double)        |
| y                     | y                                                          |        | false   | number(double)        |
| z                     | z                                                          |        | false   | number(double)        |
| yaw                   | The robot's heading angle in radians. Range: $(-\pi, \pi)$ |        | false   | number(double)        |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | mission id                                        | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": "1700000000",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# QUERY HISTORY MISSIONS BY PAGE

url: /app/openapi/v1/mission/list

method: POST

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

Note:

Example:

```
{
  "pageIndex": 0,
  "pageLength": 0,
  "requestData": {
    "deviceOpenId": "111",
    "startTimestamp": 1712725202000,
    "endTimestamp": 1712725203000,
    "isShowNavigationMission": false
  }
}
```

Params:

| name                                    | description                             | in     | require | type           |
|-----------------------------------------|-----------------------------------------|--------|---------|----------------|
| X-API-Key                               | Key                                     | header | true    | string         |
| X-API-timestamp                         | timestamp                               | header | true    | string         |
| X-API-signature                         | signature                               | header | true    | string         |
| pageQueryRequestQueryMissionOpenRequest | search by page                          | body   | true    |                |
| pageIndex                               | page index                              |        | false   | integer(int32) |
| pageLength                              | page length                             |        | false   | integer(int32) |
| requestData                             |                                         |        | true    |                |
| deviceOpenId                            | your device open id                     |        | true    | string         |
| startTimestamp                          | mission query start timestamp, unit(ms) |        | false   | integer(int64) |

| name                    | description                                                                        | in | require | type           |
|-------------------------|------------------------------------------------------------------------------------|----|---------|----------------|
| endTimeStamp            | mission query end timestamp, unit(ms), if it is not set, current date will be used |    | false   | integer(int64) |
| isShowNavigationMission | whether show navigation mission (SpecLocation or GoPoint) in result set            |    | false   | bool           |

Status:

| code | description | schema                                |
|------|-------------|---------------------------------------|
| 200  | OK          | ResponsePageDataMissionDetailOpenResp |

Response Params:

| name                  | description                                                   | type    | schema |
|-----------------------|---------------------------------------------------------------|---------|--------|
| data                  |                                                               |         |        |
| data                  |                                                               | array   |        |
| id                    | id                                                            | integer |        |
| missionId             | mission id                                                    | string  |        |
| startTime             | mission start time (formatted date based on device time zone) | string  |        |
| startTimeStamp        | mission start timestamp                                       | integer |        |
| endTime               | mission end time (formatted date based on device time zone)   | string  |        |
| endTimeStamp          | mission end timestamp                                         | integer |        |
| duration              | mission duration                                              | integer |        |
| status                | status                                                        | string  |        |
| errDesc               | the error description when the task is abnormal end           | string  |        |
| taskSummaryPictureUrl | the picture url of task summary                               | String  |        |
| totalTime             | total working time (min)                                      | integer |        |
| cleanedArea           | cleaned area (m2)                                             | integer |        |
| coverage              | coverage                                                      | string  |        |
| cleaningTime          | duration of cleaning                                          | integer |        |
| chargingTime          | duration of charging                                          | integer |        |

| name             | description                                                                                                                                                                                                                                                       | type    | schema |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| pauseTime        | duration of pause                                                                                                                                                                                                                                                 | integer |        |
| sceneName        | scene name                                                                                                                                                                                                                                                        | string  |        |
| mapVersion       | map version                                                                                                                                                                                                                                                       | string  |        |
| mapId            | map id                                                                                                                                                                                                                                                            | integer |        |
| areaList         | list of areas                                                                                                                                                                                                                                                     | array   | Area   |
| id               | user defined area id, required when creating mission/schedule(the name of the parameter is 'arealds')                                                                                                                                                             | string  |        |
| name             | area name                                                                                                                                                                                                                                                         | string  |        |
| type             | area type                                                                                                                                                                                                                                                         | string  |        |
| geo              | user defined areas                                                                                                                                                                                                                                                | array   | Point  |
| x                | x                                                                                                                                                                                                                                                                 | number  |        |
| y                | y                                                                                                                                                                                                                                                                 | number  |        |
| sweepType        | sweep type<br>- <b>SweepModeSweep</b> : Full Cleaning<br>- <b>SweepModePatrol</b> : Patrol Cleaning<br>- <b>SweepModeDeepClean</b> : Deep Cleaning<br>- <b>SweepModeSpecLocation</b> : Go to Point(x, y)<br>- <b>SweepModeGoPoint</b> : Go to Point(x, y and yaw) | string  |        |
| fixedLaps        | fixed laps                                                                                                                                                                                                                                                        | integer |        |
| deviceType       | device type, eg:'SP50'                                                                                                                                                                                                                                            | string  |        |
| water            | amount of water used in mission (L)                                                                                                                                                                                                                               | number  |        |
| cleaningSolution | amount of cleaning solution used in mission (mL)                                                                                                                                                                                                                  | integer |        |
| startBattery     | Starting battery level(relative)                                                                                                                                                                                                                                  | integer |        |
| endBattery       | Ending battery level(relative)                                                                                                                                                                                                                                    | integer |        |

| name               | description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | type           | schema   |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------|
| taskSourceTypeEnum | Task source<br>- <b>APP_TEMP</b> : Temporary task created via mobile application<br>- <b>APP_TIMING</b> : Scheduled task created via mobile application<br>- <b>APP_FAV</b> : Favorite task launched on a mobile application<br>- <b>PAD_TEMP</b> : Temporary task launched on the robot's tablet<br>- <b>PAD_TIMING</b> : Scheduled task created on the robot's tablet<br>- <b>PAD_FAV</b> : Favorite task launched on the robot's tablet<br>- <b>SCAN_CODE</b> : QR Code task<br>- <b>OPEN_API</b> : Scheduled task created via open api or temporary task launched via open api | string         |          |
| taskEndReasonEnum  | Task end reason<br>- <b>TIME_OUT</b> : Time limit reached<br>- <b>LAPS_REACHED</b> : Completed set rounds<br>- <b>APP_MANUAL</b> : Manually stopped on APP<br>- <b>PAD_MANUAL</b> : Manually stopped on PAD<br>- <b>PAD_TIMING</b> : Scheduled task created on the robot's tablet<br>- <b>SYSTEM_ABEND</b> : Favorite task launched on the robot's tablet<br>- <b>SYSTEM_PREEMPT</b> : Preempted by new task<br>- <b>JOINT_CLEAN_SRC_END</b> : Preceding sweep not finished.                                                                                                       | string         |          |
| cumulativeDistance | Distance traveled for the task, unit: meter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | integer        |          |
| meanVel            | Average speed, unit: m/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | number         |          |
| missionInitiator   | Mission Initiator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Operator |
| source             | source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | string         |          |
| user               | user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | string         |          |
| missionCanceller   | Mission Canceller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Operator |
| source             | source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | string         |          |
| user               | user                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | string         |          |
| pageIndex          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | integer(int64) |          |
| pageLength         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | integer(int64) |          |
| totalCnt           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | integer(int64) |          |
| success            | used to determine whether the call is successful                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | boolean        |          |

| name | description                                       | type           | schema |
|------|---------------------------------------------------|----------------|--------|
| code | response status code                              | integer(int32) |        |
| info | response description                              | string         |        |
| rid  | traceld is used to trace the records of API calls | string         |        |

**Response Example:**

```
{
  "data": {
    "data": [
      {
        "id": 0,
        "missionId": "",
        "startTime": "",
        "startTimeStamp": 0,
        "endTime": "",
        "endTimeStamp": 0,
        "duration": 0,
        "status": "success/canceled/error",
        "taskSummaryPictureUrl": "https://cloud-cicd-1310809401.cos.ap-
        guangzhou.myqcloud.com/screenshot_staging/task_history/52_1715598249959.webp",
        "totalTime": 0,
        "cleanedArea": 0,
        "coverage": "",
        "cleaningTime": 0,
        "chargingTime": 0,
        "pauseTime": 0,
        "sceneName": "",
        "mapVersion": "",
        "mapId": 0,
        "areaList": [
          {
            "id": "",
            "name": "",
            "type": "",
            "geo": [
              {
                "x": 0,
                "y": 0,
                "z": 0,
                "yaw": 0
              }
            ]
          }
        ]
      }
    ]
  }
}
```

```

        ],
        "sweepType": "",
        "fixedLaps": 0,
        "deviceType": "",
        "water": 0,
        "cleaningSolution": 0,
        "errDesc": "Not running for a long time",
        "startBattery": 60,
        "endBattery": 81,
        "taskSourceTypeEnum": "PAD_FAV",
        "taskEndReasonEnum": "APP_MANUAL",
        "cumulativeDistance": 123,
        "meanVel": 3.1415,
        "missionInitiator": {
            "source": "pad",
            "user": "Jane Doe"
        },
        "missionCanceller": {
            "source": "app",
            "user": "Alice"
        },
    ],
    "pageIndex": 0,
    "pageLength": 0,
    "totalCnt": 0
},
"success": true,
"code": 0,
"info": "",
"rid": ""
}

```

GO HOME

**url:** /app/openapi/v1/mission/home/{deviceOpenId}

**method:** POST

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

**Status:**

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": "",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# STOP CURRENT MISSION

**url:** `/app/openapi/v1/mission/current/stop/{deviceOpenId}`

**method:** `POST`

**produces:** `application/x-www-form-urlencoded`

**consumes:** `*/*`

**Note:**

**Params:**

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-Api-Key       | Key            | header | true    | string |        |
| X-Api-timestamp | timestamp      | header | true    | string |        |
| X-Api-signature | signature      | header | true    | string |        |

**Status:**

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# PAUSE CURRENT MISSION

url: /app/openapi/v1/mission/current/pause/{deviceOpenId}

method: POST

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name | description | type | schema |
|------|-------------|------|--------|
|------|-------------|------|--------|

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceId is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# CONTINUE CURRENT MISSION

**url:** /app/openapi/v1/mission/current/continue/{deviceOpenId}

**method:** POST

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |

| name            | description | in     | require | type   | schema |
|-----------------|-------------|--------|---------|--------|--------|
| X-Api-signature | signature   | header | true    | string |        |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

MAP

# CHANGE MAP

url: /app/openapi/v1/map/changeMap

method: POST

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": "",
  "mapId": ""
}
```

Params:

| name                 | description          | in     | require | type                 | schema               |
|----------------------|----------------------|--------|---------|----------------------|----------------------|
| X-API-Key            | Key                  | header | true    | string               |                      |
| X-API-timestamp      | timestamp            | header | true    | string               |                      |
| X-API-signature      | signature            | header | true    | string               |                      |
| changeMapOpenRequest | ChangeMapOpenRequest | body   | true    | ChangeMapOpenRequest | ChangeMapOpenRequest |
| deviceOpenId         | device open id       |        | true    | string               |                      |
| mapId                | map id               |        | true    | string               |                      |

Status:

| code | description | schema          |
|------|-------------|-----------------|
| 200  | OK          | ResponseBoolean |

Response Params:

| name    | description                                      | type    | schema |
|---------|--------------------------------------------------|---------|--------|
| data    | dataset                                          | boolean |        |
| success | used to determine whether the call is successful | boolean |        |

| name | description                                       | type           | schema         |
|------|---------------------------------------------------|----------------|----------------|
| code | response status code                              | integer(int32) | integer(int32) |
| info | response description                              | string         |                |
| rid  | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": true,
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# LIST ALL AREAS BY DEVICE

**url:** /app/openapi/v1/map/area

**method:** GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| mapId           | map id         | query  | true    | long   |        |
| deviceOpenId    | device open id | query  | true    | string |        |
| X-Api-Key       | Key            | header | true    | string |        |
| X-Api-timestamp | timestamp      | header | true    | string |        |
| X-Api-signature | signature      | header | true    | string |        |

Status:

| code | description | schema               |
|------|-------------|----------------------|
| 200  | OK          | ResponseAreaOpenResp |

Response Params:

| name     | description                                                                                           | type           | schema         |
|----------|-------------------------------------------------------------------------------------------------------|----------------|----------------|
| data     |                                                                                                       | AreaOpenResp   | AreaOpenResp   |
| mapId    | map id                                                                                                | integer(int64) |                |
| areaList | list of areas                                                                                         | array          | Area           |
| id       | user defined area id, required when creating mission/schedule(the name of the parameter is 'arealds') | string         |                |
| name     | area name                                                                                             | string         |                |
| type     | area type                                                                                             | string         |                |
| geo      | user defined areas                                                                                    | array          | Point          |
| x        | x                                                                                                     | number         |                |
| y        | y                                                                                                     | number         |                |
| success  | used to determine whether the call is successful                                                      | boolean        |                |
| code     | response status code                                                                                  | integer(int32) | integer(int32) |
| info     | response description                                                                                  | string         |                |
| rid      | traceld is used to trace the records of API calls                                                     | string         |                |

Response Example:

```
{
  "data": {
    "mapId": 0,
    "areaList": [
      {
        "id": "",
        "name": "",
        "type": "",
        "geo": [
          {
            "x": 0,
```

```
        "y": 0
      }
    ]
  },
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

## CREATE BACK POINT

**url:** /app/openapi/v1/map/backPoint

**method:** POST

**produces:** application/x-www-form-urlencoded,application/json

**consumes:** \*/\*

**Note:**

Create back point, the robot must be online, and the input point should be within the map.

**Example:**

```
{
  "deviceOpenId": "GVXCC05JPA",
  "mapId": 1721645647269112,
  "name": "bp name",
  "x": 2,
  "y": 2,
  "yaw": 1.28
}
```

**Params:**

| name      | description | in     | require | type   |
|-----------|-------------|--------|---------|--------|
| X-Api-Key | Key         | header | true    | string |

| name                       | description                                                | in     | require | type                       |
|----------------------------|------------------------------------------------------------|--------|---------|----------------------------|
| X-Api-timestamp            | timestamp                                                  | header | true    | string                     |
| X-Api-signature            | signature                                                  | header | true    | string                     |
| createBackPointOpenRequest | CreateBackPointOpenRequest                                 | body   | true    | CreateBackPointOpenRequest |
| deviceOpenId               | robot open id                                              |        | true    | string                     |
| mapId                      | map id                                                     |        | true    | integer(int64)             |
| name                       | back point name,length should less than 20                 |        | true    | string                     |
| x                          | x-coordinate                                               |        | true    | number(double)             |
| y                          | y-coordinate                                               |        | true    | number(double)             |
| yaw                        | The robot's heading angle in radians. Range: $(-\pi, \pi)$ |        | true    | number(double)             |

Status:

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | back point id                                     | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceId is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": "b87acfde-51ca-4839-897c-1b94d9159a77",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# UPDATE BACK POINT

url: /app/openapi/v1/map/backPoint

method: PUT

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

**Note:**

Update the robot's back point on a map by back point id, the robot must be online

**Example:**

```
{
  "deviceOpenId": "GVXCC05JPA",
  "mapId": 1721645647269112,
  "name": "new name",
  "x": 1,
  "y": 1,
  "yaw": 1,
  "backPointId": "b87acfde-51ca-4839-897c-1b94d9159a77"
}
```

**Params:**

| name                       | description                                                | in     | require | type           |
|----------------------------|------------------------------------------------------------|--------|---------|----------------|
| X-Api-Key                  | Key                                                        | header | true    | string         |
| X-Api-timestamp            | timestamp                                                  | header | true    | string         |
| X-Api-signature            | signature                                                  | header | true    | string         |
| updateBackPointOpenRequest | UpdateBackPointOpenRequest                                 | body   | true    |                |
| deviceOpenId               | robot open id                                              |        | true    | string         |
| mapId                      | map id                                                     |        | true    | integer(int64) |
| name                       | back point name,length should less than 20                 |        | true    | string         |
| x                          | x-coordinate                                               |        | true    | number(double) |
| y                          | y-coordinate                                               |        | true    | number(double) |
| yaw                        | The robot's heading angle in radians. Range: $(-\pi, \pi)$ |        | true    | number(double) |

| name        | description   | in | require | type   |
|-------------|---------------|----|---------|--------|
| backPointId | back point id |    | true    | string |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

LIST BACK POINTS

url: /app/openapi/v1/map/backPoint

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

**Note:**

List the robot's all back points on a map, the robot must be online

**Example:**

```
curl -H "X-API-Key:XXX" -H "X-API-timestamp:YYY" -H "X-API-signature:ZZZ" -H
"Content-Type:application/x-www-form-urlencoded" "$HOST/app/openapi/v1/map/backPoint?
deviceOpenId=GVXCC05JPA&mapId=1721645647269112"
```

**Params:**

| name            | description | in     | require | type           | schema |
|-----------------|-------------|--------|---------|----------------|--------|
| deviceOpenId    |             | query  | true    | string         |        |
| mapId           |             | query  | true    | integer(int64) |        |
| X-API-Key       | Key         | header | true    | string         |        |
| X-API-timestamp | timestamp   | header | true    | string         |        |
| X-API-signature | signature   | header | true    | string         |        |

**Status:**

| code | description | schema                   |
|------|-------------|--------------------------|
| 200  | OK          | ResponseListBackPointDTO |

**Response Params:**

| name    | description | type           | schema       |
|---------|-------------|----------------|--------------|
| data    | dataset     | array          | BackPointDTO |
| mapId   |             | integer(int64) |              |
| name    |             | string         |              |
| pointId |             | string         |              |
| pose    | back point  | Point          | Point        |
| x       | x           | number         |              |
| y       | y           | number         |              |

| name    | description                                                | type           | schema         |
|---------|------------------------------------------------------------|----------------|----------------|
| z       | z                                                          | number         |                |
| yaw     | The robot's heading angle in radians. Range: $(-\pi, \pi)$ | number(double) |                |
| success | used to determine whether the call is successful           | boolean        |                |
| code    | response status code                                       | integer(int32) | integer(int32) |
| info    | response description                                       | string         |                |
| rid     | traceId is used to trace the records of API calls          | string         |                |

**Response Example:**

```
{
  "data": [
    {
      "mapId": 1721645647269112,
      "name": "bp name",
      "pointId": "b87acfde-51ca-4839-897c-1b94d9159a77",
      "pose": {
        "x": 2,
        "y": 2,
        "yaw": 1
      }
    }
  ],
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

DELETE BACK POINT

url: /app/openapi/v1/map/backPoint

method: DELETE

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

**Note:**

Delete the robot's back point on a map by back point id, the robot must be online

**Example:**

```
{
  "deviceOpenId": "GVXCC05JPA",
  "mapId": 1721645647269112,
  "pointId": "b87acfde-51ca-4839-897c-1b94d9159a77"
}
```

**Params:**

| name                       | description                | in     | require | type           |
|----------------------------|----------------------------|--------|---------|----------------|
| X-API-Key                  | Key                        | header | true    | string         |
| X-API-timestamp            | timestamp                  | header | true    | string         |
| X-API-signature            | signature                  | header | true    | string         |
| deleteBackPointOpenRequest | DeleteBackPointOpenRequest | body   | true    |                |
| deviceOpenId               | robot open id              |        | true    | string         |
| mapId                      |                            |        | true    | integer(int64) |
| pointId                    |                            |        | true    | string         |

**Status:**

| code | description | schema         |
|------|-------------|----------------|
| 200  | OK          | ResponseString |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | string         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": "",
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

LIST ALL CURRENT MAPS OF DEVICE

url: /app/openapi/v1/map

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | query  | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

Status:

| code | description | schema                      |
|------|-------------|-----------------------------|
| 200  | OK          | ResponseListMapInfoOpenResp |

Response Params:

| name       | description                                                                                       | type           | schema          |
|------------|---------------------------------------------------------------------------------------------------|----------------|-----------------|
| data       | dataset                                                                                           | array          | MapInfoOpenResp |
| mapName    | map name                                                                                          | string         |                 |
| mapId      | map id                                                                                            | integer(int64) |                 |
| mapVersion | map version: once map is updated, map version is updated while map id and map name keep unchanged | string         |                 |
| inUse      | map is chosen and used on pad                                                                     | boolean        |                 |
| valid      | map is valid for preview                                                                          | boolean        |                 |
| success    | used to determine whether the call is successful                                                  | boolean        |                 |
| code       | response status code                                                                              | integer(int32) | integer(int32)  |
| info       | response description                                                                              | string         |                 |
| rid        | traceld is used to trace the records of API calls                                                 | string         |                 |

Response Example:

```
{
  "data": [
    {
      "mapName": "",
      "mapId": 0,
      "mapVersion": "",
      "inUse": true,
      "valid": true
    }
  ],
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

GET SCREENSHOT OF MAP

url: /app/openapi/v1/map/pic

method: GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type   | schema |
|-----------------|-------------|--------|---------|--------|--------|
| mapVersion      | map version | query  | true    | string |        |
| X-API-Key       | Key         | header | true    | string |        |
| X-API-timestamp | timestamp   | header | true    | string |        |
| X-API-signature | signature   | header | true    | string |        |

**Status:**

| code | description | schema |
|------|-------------|--------|
| 200  | OK          |        |

**Response Params:**

None

**Response Example:**

## GET SCREENSHOT OF MAP WITH FOUR COORDINATE POINTS

**url:** /app/openapi/v1/map/pic-with-points

**method:** GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type   | schema |
|-----------------|-------------|--------|---------|--------|--------|
| mapVersion      | map version | query  | true    | string |        |
| X-Api-Key       | Key         | header | true    | string |        |
| X-Api-timestamp | timestamp   | header | true    | string |        |
| X-Api-signature | signature   | header | true    | string |        |

**Status:**

| code | description | schema                 |
|------|-------------|------------------------|
| 200  | OK          | ResponseMapPicOpenResp |

**Response Params:**

| name              | description                                                                                                                                                                                              | type           | schema         |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| data              |                                                                                                                                                                                                          | MapPicOpenResp | MapPicOpenResp |
| width             | pic width                                                                                                                                                                                                | integer(int64) |                |
| height            | pic height                                                                                                                                                                                               | integer(int64) |                |
| vertexCoordinates | Four points, arrange in the following order: bottom left, top left, top right, bottom right. The coordinate aligns with the robot pose point coordinate system.<br>Value could be None if map is too old | array          | Point          |
| x                 | x                                                                                                                                                                                                        | number         |                |
| y                 | y                                                                                                                                                                                                        | number         |                |
| z                 | z                                                                                                                                                                                                        | number         |                |
| yaw               | yaw                                                                                                                                                                                                      | number         |                |
| picUrl            | url                                                                                                                                                                                                      | string         |                |
| success           | used to determine whether the call is successful                                                                                                                                                         | boolean        |                |
| code              | response status code                                                                                                                                                                                     | integer(int32) | integer(int32) |

| name | description                                       | type   | schema |
|------|---------------------------------------------------|--------|--------|
| info | response description                              | string |        |
| rid  | traceld is used to trace the records of API calls | string |        |

Response Example:

```
{
  "data": {
    "width": 0,
    "height": 0,
    "vertexCoordinates": [
      {
        "x": 0,
        "y": 0,
        "z": 0,
        "yaw": 0
      }
    ],
    "picUrl": ""
  },
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

FAVORITE

CREATE NEW FAVORITE MISSION

url: /app/openapi/v1/fav

method: POST

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": "1234",
  "timeLimit": 60,
  "fixedLaps": 0,
  "cleanEverywhere": false,
  "sweepMode": "PATROL",
  "subSweepMode": "HIGH",
  "edgeMode": "ONLY",
  "windMode": "LOW",
  "areaIds": [
    "0",
    "3"
  ],
  "userDefinedArea": [
    {
      "x": 0,
      "y": 1,
      "z": 0
    },
    {
      "x": 0,
      "y": 2,
      "z": 0
    },
    {
      "x": 1,
      "y": 1,
      "z": 0
    }
  ],
  "cleanAreasInOrder": true,
  "missionName": "",
  "backPointId": "",
  "mapId": 0
}
```

Params:

| name      | description | in     | require | type   |
|-----------|-------------|--------|---------|--------|
| X-API-Key | Key         | header | true    | string |

| name                      | description                                                                                                                                                        | in     | require | type           |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|----------------|
| X-Api-timestamp           | timestamp                                                                                                                                                          | header | true    | string         |
| X-Api-signature           | signature                                                                                                                                                          | header | true    | string         |
| createFavoriteOpenRequest | CreateFavoriteOpenRequest                                                                                                                                          | body   | true    |                |
| deviceOpenId              | your device open id                                                                                                                                                |        | true    | string         |
| timeLimit                 | task time limit, unit(minutes), (1, 1440). Will be ignored if HOMING                                                                                               |        | false   | integer(int32) |
| fixedLaps                 | fixed laps, required when duration is null, (1,3). Will be ignored if HOMING                                                                                       |        | false   | integer(int32) |
| cleanEveryWhere           | option to enable/disable cleaning of the entire place. if false,arealds should not empty. Will be ignored if HOMING                                                |        | false   | boolean        |
| sweepMode                 | only PATROL and SWEEP can allow in the favorite,Availble:PATROL,HOMING,SWEEP                                                                                       |        | true    | string         |
| subSweepMode              | sweep intensity,Availble:VERY_QUITE,LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                     |        | false   | string         |
| edgeMode                  | edge cleaning(ONLY/OFF/ON),Availble:ONLY,OFF,ON                                                                                                                    |        | false   | string         |
| windMode                  | Absorption strength(LOW/MEDIUM/HIGH),Availble:LOW,MEDIUM,HIGH                                                                                                      |        | false   | string         |
| arealds                   | clean areas, required when cleanEveryWhere is false                                                                                                                |        | false   | array          |
| userDefinedArea           | user defined areas                                                                                                                                                 |        | false   | array          |
| x                         | x                                                                                                                                                                  |        | false   | number         |
| y                         | y                                                                                                                                                                  |        | false   | number         |
| z                         | z                                                                                                                                                                  |        | false   | number         |
| yaw                       | The robot's heading angle in radians. Range: $(-\pi, \pi)$                                                                                                         |        | false   | number(double) |
| cleanAreasInOrder         | Whether to clean in the order of the area                                                                                                                          |        | false   | boolean        |
| missionName               | mission name                                                                                                                                                       |        | true    | string         |
| backPointId               | 1. null or empty string is default behavior go back to station; 2. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint |        | false   | string         |
| mapId                     | map id                                                                                                                                                             |        | true    | integer(int64) |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

## UPDATE FAVORITE MISSION

**url:** `/app/openapi/v1/fav`

**method:** `PUT`

**produces:** `application/x-www-form-urlencoded,application/json`

**consumes:** `*/*`

**Note:**

**Example:**

```
{
  "deviceOpenId": "1234",
  "timeLimit": 60,
  "fixedLaps": 0,
  "cleanEverywhere": false,
  "sweepMode": "PATROL",
  "subSweepMode": "HIGH",
  "edgeMode": "ONLY",
  "windMode": "LOW",
  "areaIds": [
    "0",
    "3"
  ],
  "userDefinedArea": [
    {
      "x": 0,
      "y": 1,
      "z": 0
    },
    {
```

```
        "x": 0,
        "y": 2,
        "z": 0
    },
    {
        "x": 1,
        "y": 1,
        "z": 0
    }
],
"cleanAreasInOrder": true,
"missionName": "",
"backPointId": "id",
"missionId": ""
}
```

Params:

| name                      | description                                                                                                                                                       | in     | require | type           | schema |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|----------------|--------|
| X-Api-Key                 | Key                                                                                                                                                               | header | true    | string         |        |
| X-Api-timestamp           | timestamp                                                                                                                                                         | header | true    | string         |        |
| X-Api-signature           | signature                                                                                                                                                         | header | true    | string         |        |
| updateFavoriteOpenRequest | UpdateFavoriteOpenRequest                                                                                                                                         | body   | true    |                |        |
| deviceOpenId              | your device open id                                                                                                                                               |        | true    | string         |        |
| timeLimit                 | task time limit, unit(minutes), (1, 1440). Will be ignored if HOMING                                                                                              |        | false   | integer(int32) |        |
| fixedLaps                 | fixed laps, required when duration is null, (1,3). Will be ignored if HOMING                                                                                      |        | false   | integer(int32) |        |
| cleanEveryWhere           | option to enable/disable cleaning of the entire place. if false,arealds should not empty. Will be ignored if HOMING                                               |        | false   | boolean        |        |
| sweepMode                 | only PATROL and SWEEP can allow in the favorite,Avalible:PATROL,HOMING,SWEEP                                                                                      |        | true    | string         |        |
| subSweepMode              | sweep intensity,Avalible:VERY_QUITE,LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                    |        | false   | string         |        |
| edgeMode                  | edge cleaning(ONLY/OFF/ON),Avalible:ONLY,OFF,ON                                                                                                                   |        | false   | string         |        |
| windMode                  | Absorption strength(LOW/MEDIUM/HIGH)),Avalible:LOW,MEDIUM,HIGH                                                                                                    |        | false   | string         |        |
| arealds                   | clean areas, required when cleanEveryWhere is false                                                                                                               |        | false   | array          | string |
| userDefinedArea           | user defined areas                                                                                                                                                |        | false   | array          | Point  |
| x                         | x                                                                                                                                                                 |        | false   | number         |        |
| y                         | y                                                                                                                                                                 |        | false   | number         |        |
| z                         | z                                                                                                                                                                 |        | false   | number         |        |
| yaw                       | The robot's heading angle in radians. Range: $(-\pi, \pi)$                                                                                                        |        | false   | number(double) |        |
| cleanAreasInOrder         | Whether to clean in the order of the area                                                                                                                         |        | false   | boolean        |        |
| missionName               | mission name                                                                                                                                                      |        | true    | string         |        |
| backPointId               | 1. null or empty string is default behavior go back to station 2. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint |        | false   | string         |        |
| missionId                 | mission id                                                                                                                                                        |        | true    | string         |        |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

DELETE FAVORITE MISSION

url: /app/openapi/v1/fav

method: DELETE

produces: application/x-www-form-urlencoded,application/json

consumes: \*/\*

Note:

Example:

```
{
  "deviceOpenId": "1234",
  "missionId": ""
}
```

Params:

| name                    | description             | in     | require | type                    | schema                  |
|-------------------------|-------------------------|--------|---------|-------------------------|-------------------------|
| X-Api-Key               | Key                     | header | true    | string                  |                         |
| X-Api-timestamp         | timestamp               | header | true    | string                  |                         |
| X-Api-signature         | signature               | header | true    | string                  |                         |
| favoriteLiteOpenRequest | FavoriteLiteOpenRequest | body   | true    | FavoriteLiteOpenRequest | FavoriteLiteOpenRequest |
| deviceOpenId            | your device open id     |        | true    | string                  |                         |
| missionId               | mission id              |        | false   | string                  |                         |

Status:

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

Response Params:

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# RUN FAVORITE MISSION

**url:** /app/openapi/v1/fav/run

**method:** POST

**produces:** application/x-www-form-urlencoded, application/json

**consumes:** \*/\*

**Note:**

**Example:**

```
{
  "deviceOpenId": "1234",
  "missionId": ""
}
```

**Params:**

| name                    | description             | in     | require | type                    | schema                  |
|-------------------------|-------------------------|--------|---------|-------------------------|-------------------------|
| X-API-Key               | Key                     | header | true    | string                  |                         |
| X-API-timestamp         | timestamp               | header | true    | string                  |                         |
| X-API-signature         | signature               | header | true    | string                  |                         |
| favoriteLiteOpenRequest | FavoriteLiteOpenRequest | body   | true    | FavoriteLiteOpenRequest | FavoriteLiteOpenRequest |
| deviceOpenId            | your device open id     |        | true    | string                  |                         |
| missionId               | mission id              |        | false   | string                  |                         |

**Status:**

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

**Response Params:**

| name    | description                                      | type    | schema |
|---------|--------------------------------------------------|---------|--------|
| data    | dataset                                          | object  |        |
| success | used to determine whether the call is successful | boolean |        |

| name | description                                       | type           | schema         |
|------|---------------------------------------------------|----------------|----------------|
| code | response status code                              | integer(int32) | integer(int32) |
| info | response description                              | string         |                |
| rid  | traceId is used to trace the records of API calls | string         |                |

**Response Example:**

```

{
  "data": {},
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# LIST FAVORITE MISSIONS

**url:** /app/openapi/v1/fav/list

**method:** GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type           | schema |
|-----------------|-------------|--------|---------|----------------|--------|
| deviceOpenId    |             | query  | true    | string         |        |
| X-API-Key       | Key         | header | true    | string         |        |
| X-API-timestamp | timestamp   | header | true    | string         |        |
| X-API-signature | signature   | header | true    | string         |        |
| mapId           |             | query  | false   | integer(int64) |        |

Status:

| code | description | schema                       |
|------|-------------|------------------------------|
| 200  | OK          | ResponseListFavoriteOpenResp |

Response Params:

| name              | description                                                                                                                                                        | type           | schema           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|
| data              | dataset                                                                                                                                                            | array          | FavoriteOpenResp |
| missionId         | mission id                                                                                                                                                         | string         |                  |
| mapId             | map id                                                                                                                                                             | integer(int64) |                  |
| sceneName         | scene name                                                                                                                                                         | string         |                  |
| needModify        | if the map is deleted in the mission, the mission can be blocked                                                                                                   | boolean        |                  |
| reason            | reasons to block mission execute                                                                                                                                   | string         |                  |
| cleanEveryWhere   | option to enable/disable cleaning of the entire place. if false,arealds should not empty. Will be ignored if HOMING                                                | boolean        |                  |
| sweepMode         | clean mode,Available:PATROL,HOMING,SWEEP                                                                                                                           | string         |                  |
| subSweepMode      | sweep intensity,Available:VERY_QUIET,LOW,MEDIUM,HIGH,SPOT_CLEAN                                                                                                    | string         |                  |
| edgeMode          | edge cleaning(ONLY/OFF/ON),Available:ONLY,OFF,ON                                                                                                                   | string         |                  |
| windMode          | absorption strength(LOW/MEDIUM/HIGH)),Available:LOW,MEDIUM,HIGH                                                                                                    | string         |                  |
| arealds           | clean areas, required when cleanEveryWhere is false                                                                                                                | array          | string           |
| userDefinedArea   | user defined areas                                                                                                                                                 | array          | Point            |
| x                 | x                                                                                                                                                                  | number         |                  |
| y                 | y                                                                                                                                                                  | number         |                  |
| missionName       | mission name                                                                                                                                                       | string         |                  |
| backPointId       | 1. null or empty string is default behavior go back to station; 2. a valid back point id specify a customize back point. created by POST /openapi/v1/map/backPoint | string         |                  |
| backPoint         |                                                                                                                                                                    | Point          | Point            |
| x                 | x                                                                                                                                                                  | number         |                  |
| y                 | y                                                                                                                                                                  | number         |                  |
| z                 | z                                                                                                                                                                  | number         |                  |
| yaw               | The robot's heading angle in radians. Range: $(-\pi, \pi)$                                                                                                         | number(double) |                  |
| timeLimit         | time limit                                                                                                                                                         | integer(int64) |                  |
| newAreaDisplay    | area display                                                                                                                                                       | string         |                  |
| cleanAreasInOrder | Whether to clean in the order of the area                                                                                                                          | boolean        |                  |
| fixedLaps         | fixed laps, required when duration is null, (1,3). Will be ignored if HOMING                                                                                       | integer(int64) |                  |
| success           | used to determine whether the call is successful                                                                                                                   | boolean        |                  |
| code              | response status code                                                                                                                                               | integer(int32) | integer(int32)   |
| info              | response description                                                                                                                                               | string         |                  |
| rid               | traceld is used to trace the records of API calls                                                                                                                  | string         |                  |

Response Example:

```

{
  "data": [
    {
      "missionId": "",
      "mapId": 0,
      "sceneName": "",
      "needModify": true,
      "reason": "",
      "cleanEverywhere": false,
      "sweepMode": "PATROL",
      "subSweepMode": "HIGH",
      "edgeMode": "ONLY",
      "windMode": "LOW",
      "areaIds": "[\\"0\\",\\"3\\"]",
      "userDefinedArea": "[{\\"x\\":0,\\"y\\":1,\\"z\\":0},{\\"x\\":0,\\"y\\":2,\\"z\\":0},
{\\"x\\":1,\\"y\\":1,\\"z\\":0}]",
      "missionName": "",
      "backPointId": "-1",
      "backPoint": {
        "x": 0,
        "y": 0,
        "z": 0,
        "yaw": 0
      },
      "timeLimit": 0,
      "newAreaDisplay": "",
      "cleanAreasInOrder": true,
      "fixedLaps": 0
    }
  ],
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}

```

DEVICE

# GET DEVICE STATUS

url: /app/openapi/v1/device/status/{deviceOpenId}

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceOpenId    | device open id | path   | true    | string |        |
| X-Api-Key       | Key            | header | true    | string |        |
| X-Api-timestamp | timestamp      | header | true    | string |        |
| X-Api-signature | signature      | header | true    | string |        |

Status:

| code | description | schema                             |
|------|-------------|------------------------------------|
| 200  | OK          | ResponseDeviceStatusDetailOpenResp |

Response Params:

| name              | description                                                       | type           | schema |
|-------------------|-------------------------------------------------------------------|----------------|--------|
| data              |                                                                   |                |        |
| online            | whether the device is online                                      | boolean        |        |
| soc               | remaining battery power                                           | integer(int32) |        |
| running           | whether the device is operating                                   | boolean        |        |
| charging          | whether the device is charging                                    | boolean        |        |
| chargingMode      | charging mode, AUTO or MANUAL, values:AUTO,MANUAL,PROBABLY_MANUAL | string         |        |
| currentMapVersion | current map version                                               | string         |        |
| currentMapId      | current map id                                                    | integer(int64) |        |

| name                  | description                                                                                                                                                         | type    | schema         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------|
| currentMapName        | current map name                                                                                                                                                    | string  |                |
| vehicleTime           | last online time of device                                                                                                                                          | string  |                |
| licensePlate          | licensePlate                                                                                                                                                        | string  |                |
| remark                | robot remark on Operation platform                                                                                                                                  | string  |                |
| isEmergency           | whether the emergency mode is enabled                                                                                                                               | boolean |                |
| isManual              | whether the manual mode is enabled                                                                                                                                  | boolean |                |
| pose                  | posture signifies different meanings offline and online. online, it denotes the current real-time position, and offline implies the most recently reported position | Pose    | Pose           |
| x                     | x                                                                                                                                                                   | number  |                |
| y                     | y                                                                                                                                                                   | number  |                |
| yaw                   | The robot's heading angle in radians. Range: $(-\pi, \pi)$                                                                                                          | number  |                |
| dirtyBox              | dirty water level                                                                                                                                                   | array   | integer(int32) |
| waterBox              | clean water level                                                                                                                                                   | array   | integer(int32) |
| dockingStationDetail  | docking station detail                                                                                                                                              |         |                |
| isDocked              | is docked                                                                                                                                                           | boolean |                |
| type                  | docking station type,values:CCS_01,CCS_02,CWS_01                                                                                                                    | string  |                |
| dirtyWaterStatus      | dirty water status,values:IDLE,DRAINING_SEWAGE,FINISHED,ERROR                                                                                                       | string  |                |
| cleanWaterStatus      | clean water status,values:IDLE,ADDING_WATER,ADDING_CLEANING_SOLUTION,FINISHED,ERROR                                                                                 | string  |                |
| missionTaskDetail     | mission task detail                                                                                                                                                 |         |                |
| currentMissionId      | mission's id that device is running                                                                                                                                 | string  |                |
| sweepMode             | current task sweep mode, values:NoTask,Sweep,Patrol,Charge,Homing,ReplaceWater                                                                                      | string  |                |
| taskStatus            | current task status,values:READY,ON_GOING,PAUSE,DONE,STOP,ROUTING,ABEND,CHARGING,ABSORBS_WATER,BACK_TO_STATION,MANUAL_HANDLING,CHARGING_BROKEN,WARNING,UNKNOWN      | string  |                |
| taskProgress          | task progress                                                                                                                                                       | number  |                |
| cleanedAreaSize       | cleaned area size(m²)                                                                                                                                               | number  |                |
| remainingAreaSize     | remaining area size(m²)                                                                                                                                             | number  |                |
| remainingTime         | remaining time(second)                                                                                                                                              | integer |                |
| currentLap            | current lap                                                                                                                                                         | integer |                |
| currentAreaIds        | current area ids                                                                                                                                                    | array   | string         |
| consumablesStatistics | consumables statistics, only for SP50                                                                                                                               | array   |                |
| name                  | consumables name                                                                                                                                                    | string  |                |
| validityTime          | duration of validity (hour)                                                                                                                                         | integer |                |
| remainingPercentage   | remaining percentage                                                                                                                                                | string  |                |
| success               | used to determine whether the call is successful                                                                                                                    | boolean |                |

| name | description                                       | type           | schema         |
|------|---------------------------------------------------|----------------|----------------|
| code | response status code                              | integer(int32) | integer(int32) |
| info | response description                              | string         |                |
| rid  | traceld is used to trace the records of API calls | string         |                |

Response Example:

```
{
  "data": {
    "online": true,
    "soc": 0,
    "running": true,
    "charging": true,
    "chargingMode": "",
    "currentMapVersion": "",
    "currentMapId": 0,
    "currentMapName": "",
    "vehicleTime": "",
    "licensePlate": "",
    "remark": "User Defined Remark",
    "isEmergency": true,
    "isManual": true,
    "pose": {
      "x": 0,
      "y": 0,
      "yaw": 0
    },
    "dirtyBox": [],
    "waterBox": [],
    "dockingStationDetail": {
      "isDocked": true,
      "type": "",
      "dirtyWaterStatus": "",
      "cleanWaterStatus": ""
    },
    "missionTaskDetail": {
      "currentMissionId": "",
      "sweepMode": "",
      "taskStatus": "",
      "taskProgress": 0,
      "cleanedAreaSize": 0,
      "remainingAreaSize": 0,
      "remainingTime": 0
    },
    "consumablesStatistics": [
      {
```

```
        "name": "",
        "validityTime": 0,
        "remainingPercentage": ""
      }
    ]
  },
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# GET ROBOT INFO

url: /app/openapi/v1/device/info/{deviceId}

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   |
|-----------------|----------------|--------|---------|--------|
| deviceId        | device open id | path   | true    | string |
| X-API-Key       | Key            | header | true    | string |
| X-API-timestamp | timestamp      | header | true    | string |
| X-API-signature | signature      | header | true    | string |

Status: open.md

| code | description | schema                     |
|------|-------------|----------------------------|
| 200  | OK          | ResponseDeviceInfoOpenResp |

Response Params:

| name              | description                                       | type               |
|-------------------|---------------------------------------------------|--------------------|
| data              |                                                   | DeviceInfoOpenResp |
| licensePlate      | licensePlate                                      | string             |
| remark            | robot remark on Operation platform                | string             |
| buildingName      | building name                                     | string             |
| serialNumber      | serial number                                     | string             |
| softwareVersion   | software version                                  | string             |
| activated         | activated or not                                  | boolean            |
| activateTimestamp | active timestamp                                  | integer(int64)     |
| success           | used to determine whether the call is successful  | boolean            |
| code              | response status code                              | integer(int32)     |
| info              | response description                              | string             |
| rid               | traceld is used to trace the records of API calls | string             |

Response Example:

```
{
  "data": {
    "licensePlate": "",
    "remark": "",
    "buildingName": "",
    "serialNumber": "",
    "softwareVersion": "",
    "activated": true,
    "activateTimestamp": 0
  },
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

# GET ROBOT'S MAINTENANCE DETAIL

url: /app/openapi/v1/device/maintenance-detail/{deviceOpenId}

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description | in     | require | type   |
|-----------------|-------------|--------|---------|--------|
| deviceOpenId    |             | path   | true    | string |
| X-Api-Key       | Key         | header | true    | string |
| X-Api-timestamp | timestamp   | header | true    | string |
| X-Api-signature | signature   | header | true    | string |

Status:

| code | description | schema                            |
|------|-------------|-----------------------------------|
| 200  | OK          | ResponseMaintenanceDetailOpenResp |

Response Params:

| name             | description                                  | type                      |
|------------------|----------------------------------------------|---------------------------|
| data             |                                              | MaintenanceDetailOpenResp |
| maintenanceltems | Consumables                                  | array                     |
| name             | Consumable Name                              | string                    |
| remainPercent    | Remaining Life Percentage                    | string                    |
| remainHours      | Shows Est. Remaining Hours if life remains   | integer                   |
| overDueHours     | Shows Overdue Hours if consumable is overdue | integer                   |

| name           | description                                          | type           |
|----------------|------------------------------------------------------|----------------|
| life           | Consumable life                                      | integer        |
| history        | Maintenance History                                  | array          |
| resetTimestamp | Reset timestamp                                      | integer        |
| resetTime      | String format of reset time at the robot's time zone | string         |
| user           | Reset user                                           | string         |
| source         | Reset source                                         | string         |
| success        | used to determine whether the call is successful     | boolean        |
| code           | response status code                                 | integer(int32) |
| info           | response description                                 | string         |
| rid            | traceld is used to trace the records of API calls    |                |

**Response Example:**

```
{
  "code": 200,
  "data": {
    "maintenanceItems": [
      {
        "name": "Squeegee Blade",
        "history": [
          {
            "resetTimestamp": 1772586706000,
            "source": "pad",
            "resetTime": "Wed, Mar 04, 2026 9:11",
            "user": "Admin"
          },
          {
            "resetTimestamp": 1772586704000,
            "source": "pad",
            "resetTime": "Wed, Mar 04, 2026 9:11",
            "user": "Admin"
          }
        ]
      },
      {
        "resetTimestamp": 1772586704000,
        "source": "pad",
        "resetTime": "Wed, Mar 04, 2026 9:11",
        "user": "Admin"
      }
    ],
    "overDueHours": 2,
    "life": 150,
    "remainPercent": "0%"
  }
}
```

```
        "remainHours": 480,
        "name": "Detergent",
        "history": [
            {
                "resetTimestamp": 1743144231000,
                "source": "pad",
                "resetTime": "Fri, Mar 28, 2025 14:43",
                "user": "Admin"
            }
        ],
        "life": 480,
        "remainPercent": "100%"
    }
]
},
"success": true,
"rid": "ec8edcf921e041a09054b8ac28f3479d.668.17750218761290019"
}
```

## GET ROBOT SETTING

**url:** /app/openapi/v1/device/setting/{deviceOpenId}

**method:** GET

**produces:** application/x-www-form-urlencoded

**consumes:** \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type   | schema |
|-----------------|-------------|--------|---------|--------|--------|
| deviceOpenId    |             | path   | true    | string |        |
| X-API-Key       | Key         | header | true    | string |        |
| X-API-timestamp | timestamp   | header | true    | string |        |
| X-API-signature | signature   | header | true    | string |        |

Status:

| code | description | schema                   |
|------|-------------|--------------------------|
| 200  | OK          | ResponseRobotSettingData |

Response Params:

| name     | description                                       | type             | schema           |
|----------|---------------------------------------------------|------------------|------------------|
| data     |                                                   | RobotSettingData | RobotSettingData |
| runDebug | Is the debug mode enabled for the robot           | boolean          |                  |
| success  | used to determine whether the call is successful  | boolean          |                  |
| code     | response status code                              | integer(int32)   | integer(int32)   |
| info     | response description                              | string           |                  |
| rid      | traceld is used to trace the records of API calls | string           |                  |

Response Example:

```
{
  "data": {
    "runDebug": true
  },
  "success": true,
  "code": 200,
  "info": "",
  "rid": ""
}
```

UPDATE ROBOT SETTING

url: /app/openapi/v1/device/setting/{deviceOpenId}

method: PUT

produces: application/x-www-form-urlencoded, application/json

consumes: \*/\*

**Note:**

**Example:**

```
{
  "runDebug": true
}
```

**Params:**

| name                      | description                          | in     | require | type    |
|---------------------------|--------------------------------------|--------|---------|---------|
| deviceOpenId              |                                      | path   | true    | string  |
| X-API-Key                 | Key                                  | header | true    | string  |
| X-API-timestamp           | timestamp                            | header | true    | string  |
| X-API-signature           | signature                            | header | true    | string  |
| updateRobotSettingOpenReq | UpdateRobotSettingOpenReq            | body   | true    |         |
| runDebug                  | true - enable the robot's debug mode |        | false   | boolean |

**Status:**

| code | description | schema       |
|------|-------------|--------------|
| 200  | OK          | ResponseVoid |

**Response Params:**

| name    | description                                       | type           | schema         |
|---------|---------------------------------------------------|----------------|----------------|
| data    | dataset                                           | object         |                |
| success | used to determine whether the call is successful  | boolean        |                |
| code    | response status code                              | integer(int32) | integer(int32) |
| info    | response description                              | string         |                |
| rid     | traceld is used to trace the records of API calls | string         |                |

**Response Example:**

```
{
  "data": null,
  "success": true,
  "code": 200,
  "info": "",
  "rid": ""
}
```

# LIST DEVICE SYSTEM ERRORS

url: /app/openapi/v1/device/{deviceId}/sys-error/list

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

Note:

Params:

| name            | description    | in     | require | type   | schema |
|-----------------|----------------|--------|---------|--------|--------|
| deviceId        | device open id | path   | true    | string |        |
| X-API-Key       | Key            | header | true    | string |        |
| X-API-timestamp | timestamp      | header | true    | string |        |
| X-API-signature | signature      | header | true    | string |        |

Status:

| code | description | schema                          |
|------|-------------|---------------------------------|
| 200  | OK          | ResponseListSystemErrorOpenResp |

Response Params:

| name | description | type | schema |
|------|-------------|------|--------|
|------|-------------|------|--------|

| name        | description                                       | type           | schema              |
|-------------|---------------------------------------------------|----------------|---------------------|
| data        | dataset                                           | array          | SystemErrorOpenResp |
| code        |                                                   | integer(int32) |                     |
| name        |                                                   | string         |                     |
| description |                                                   | string         |                     |
| success     | used to determine whether the call is successful  | boolean        |                     |
| code        | response status code                              | integer(int32) | integer(int32)      |
| info        | response description                              | string         |                     |
| rid         | traceld is used to trace the records of API calls | string         |                     |

Response Example:

```
{
  "data": [
    {
      "code": 703903,
      "name": "Charging Failed",
      "description": "Charging Failed: Outbound adjustment failed in precise
recharge operation"
    }
  ],
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```

GET DEVICE OPEN-ID LIST

url: /app/openapi/v1/device/deviceOpenIds

method: GET

produces: application/x-www-form-urlencoded

consumes: \*/\*

**Note:**

**Params:**

| name            | description | in     | require | type   | schema |
|-----------------|-------------|--------|---------|--------|--------|
| X-Api-Key       | Key         | header | true    | string |        |
| X-Api-timestamp | timestamp   | header | true    | string |        |
| X-Api-signature | signature   | header | true    | string |        |

**Status:**

| code | description | schema                       |
|------|-------------|------------------------------|
| 200  | OK          | ResponseListDeviceOpenIdResp |

**Response Params:**

| name         | description                                       | type           | schema           |
|--------------|---------------------------------------------------|----------------|------------------|
| data         | dataset                                           | array          | DeviceOpenIdResp |
| licensePlate | license plate                                     | string         |                  |
| deviceOpenId | device open id                                    | string         |                  |
| success      | used to determine whether the call is successful  | boolean        |                  |
| code         | response status code                              | integer(int32) | integer(int32)   |
| info         | response description                              | string         |                  |
| rid          | traceld is used to trace the records of API calls | string         |                  |

**Response Example:**

```
{
  "data": [
    {
      "licensePlate": "",
      "deviceOpenId": ""
    }
  ],
  "success": true,
  "code": 0,
  "info": "",
  "rid": ""
}
```